

SOFTWARE CHANGE IMPLEMENTATION REPORT

Student Management System

Change Request: CR-004
Student Input Field Validation

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Student ID: 24ICT06

Software Configuration Management - SCM (K72C001M12)
Version 1.0

08 October 2026

University College of Jaffna / Department of Interdisciplinary Studies

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1. Change Request

1.1 Change Request Identification

Field	Details
Change Request ID	CR-004
Change Request Title	Student Input Field Validation
Priority	Not specified in the supplied implementation report
Difficulty	Not specified in the supplied implementation report
Change Type	Enhancement / Frontend and Backend Validation Change
Implementation Status	Completed
Student / Developer	C. Lavasujan (Student ID: 24ICT06)
Date	08 October 2026

1.2 Business Requirement

Student creation and editing must enforce consistent validation so that incomplete, malformed, insecure, or whitespace-padded values are rejected before being stored or processed. Validation must provide clear user feedback in the StudentFormModal while also protecting the backend against requests that bypass the frontend.

The change is intended to improve data quality and reduce validation-bypass vulnerabilities without disrupting the existing student CRUD operations, administrator authentication, authorization, or password hashing behaviour.

1.3 Existing Behaviour

Prior to implementing this change request, the student Add and Edit workflow required stronger validation controls than those consistently enforced at both client and server layers. The requested implementation therefore adds explicit validation to the React form and independently re-validates the same constraints in the Express/Node.js controller.

- Client-side validation was not sufficient to guarantee protection against direct or simulated API requests.
- Student name fields require length and alphabetic-content validation.
- Email and phone values require format and length validation.
- New student passwords require a defined strong-password policy.
- Whitespace padding must not be usable to circumvent required-field rules.

1.4 Requested Change

Implement comprehensive client-side and server-side validation for the student Add and Edit operations. The frontend must prevent invalid submissions and display precise messages, while the backend must independently enforce all required rules and return HTTP 400 Bad Request for invalid requests.

- Registration Number must be provided and must not be entirely blank or whitespace.
- First Name must be provided, contain at least 3 characters, and contain only alphabetic characters.
- Last Name must be provided, contain at least 3 characters, and contain only alphabetic characters.
- Email must be provided and follow a standard email format.
- Phone Number must be provided, numeric, and exactly 10 digits long.
- For new student creation, Password must be non-empty, at least 8 characters, and contain uppercase, lowercase, number, and special-character content.
- All string-based inputs must be trimmed before validation checks are applied.
- Invalid client-side input must be rejected before an API request is attempted.
- Backend addStudent and editStudent endpoints must re-verify all validation rules and return HTTP 400 before database operations when validation fails.
- Existing Edit, Delete, authentication, authorization, and password hashing mechanisms must remain functional.

1.5 Change Objective

The objective of CR-004 is to establish consistent and authoritative validation for student data across the frontend and backend, provide clear validation feedback, prevent client-side bypasses, and preserve existing CRUD and security behaviour.

2. Change Analysis & Impact

2.1 Change Scope

In Scope

- Client-side validation within StudentFormModal.jsx for student Add/Edit submission.
- Server-side validation within studentController.js for addStudent and editStudent.
- Validation and error messaging for registration number, first name, last name, email, phone number, and new-student password.
- Whitespace trimming before validation.
- HTTP 400 Bad Request responses for invalid direct API requests.
- Manual and API verification of valid creation, valid update, invalid requests, and preservation of Edit/Delete functionality.

Out of Scope

- Database schema redesign or new database columns.
- Modification of studentModel.js SQL queries.
- Changes to routing or dashboard data-fetching pipelines.
- Redesign of authentication, authorization, or password hashing mechanisms.
- Unrelated student-management features.

2.2 Assumptions and Constraints

Assumptions

- The existing student data model and CRUD operations remain the underlying persistence mechanism.
- The existing authentication and authorization middleware continues to supply the required request context.
- The StudentFormModal remains the primary client-side Add/Edit form.

Constraints

- Validation rules must be applied consistently in both client and server layers.
- Invalid API requests must be rejected before database operations are attempted.
- The change must not alter existing student identifiers or CRUD targeting.
- Password rules apply to new student creation; existing password hashing remains unchanged.

2.3 Impact Analysis

Identify the areas that could be affected by the change.

Area	Impact	Details
Frontend	High	StudentFormModal validation and error-state handling change the Add/Edit submission path and may affect form usability.

Area	Impact	Details
Backend	High	studentController.js independently validates all fields and introduces authoritative HTTP 400 rejection behaviour.
Database	Low	No schema or SQL model changes are required; database operations are protected by validation before processing.
API	High	Existing student create/edit endpoints now reject invalid payloads with descriptive 400 responses.
Authentication / Authorization	Low	Existing access-control behaviour is preserved without changing the authentication middleware.
Security	High	Server-side re-validation prevents frontend validation bypass and whitespace circumvention.
User Interface / Styling	Medium	formError messages must remain visible and clear inside StudentFormModal.
Configuration	None	No configuration or environment-variable changes are specified.
External Services	None	No third-party APIs or external services are affected.
Performance	Low	Validation is lightweight and occurs before database work for invalid requests.
Testing	High	Client, server, security, CRUD, regression, and API verification are required.
Documentation	Medium	Implementation, acceptance, evidence, and traceability records are required.

2.4 Affected Components and Files

Component / File	Type	Changed?	Purpose / Reason for Change
frontend/src/components/StudentFormModal.jsx	Frontend	Yes	Added client-side validation, trimming, and precise formError messages.
backend/src/controllers/studentController.js	Backend / API	Yes	Added authoritative validation to addStudent and editStudent; invalid requests return HTTP 400.
backend/src/models/studentModel.js	Backend Model	No	Existing SQL and student persistence logic remain unchanged.
Database Schema	Database	No	Database schema redesign is outside the scope of CR-004.
Student Dashboard / CRUD Flow	Frontend / Integration	Regression Verified	Existing Add, Edit, and Delete targeting must remain correct.

3. Requirements & Acceptance Criteria

3.1 Functional Requirements

ID	Requirement	Status
FR-01	Registration Number must be provided and not entirely empty or whitespace.	Met
FR-02	First Name must be provided, at least 3 characters long, and alphabetic only.	Met
FR-03	Last Name must be provided, at least 3 characters long, and alphabetic only.	Met
FR-04	Email must be provided and follow the required email format.	Met
FR-05	Phone Number must be provided, numeric only, and exactly 10 digits.	Met
FR-06	New student Password must not be empty.	Met
FR-07	New student Password must be at least 8 characters long.	Met
FR-08	New student Password must include uppercase, lowercase, number, and special-character content.	Met
FR-09	The same validation rules must be enforced consistently in frontend and backend layers.	Met
FR-10	All string inputs must be trimmed before validation.	Met
FR-11	Invalid frontend submissions must be prevented before an API request is made.	Met
FR-12	Backend validation failures must return HTTP 400 Bad Request with descriptive feedback.	Met
FR-13	Valid student creation must succeed when all required values satisfy the rules.	Met
FR-14	Valid student update must succeed when all edited values satisfy the rules.	Met
FR-15	Invalid Add and Edit submissions must be rejected rather than persisted.	Met
FR-16	Existing Edit and Delete actions must continue to target the correct student_id.	Met
FR-17	Existing administrator authentication and authorization behaviour must remain operational.	Met
FR-18	Existing password hashing and student CRUD mechanisms must remain unchanged except for validation enforcement.	Met

3.2 Non-Functional Requirements

ID	Requirement	Status
NFR-01	Reliability: Invalid data must be rejected predictably at both client and server layers.	Met
NFR-02	Usability: Validation messages must be specific and understandable to the	Met

ID	Requirement	Status
	administrator.	
NFR-03	Data Integrity: Invalid requests must be rejected before database processing.	Met
NFR-04	Backward Compatibility: Existing CRUD, authentication, authorization, and password hashing behaviour remains intact.	Met

3.3 Security Requirements

ID	Requirement	Status
SEC-001	Server-side validation must remain authoritative and must not rely solely on frontend checks.	Met
SEC-002	The addStudent endpoint must independently validate incoming student data.	Met
SEC-003	The editStudent endpoint must independently validate incoming student data.	Met
SEC-004	Invalid requests must be rejected with HTTP 400 before database operations are attempted.	Met
SEC-005	String trimming must prevent whitespace padding from circumventing required-field checks.	Met
SEC-006	Existing authentication and authorization protections must not be bypassed by the validation change.	Met

3.4 Acceptance Criteria

ID	Acceptance Criterion	Verification Method	Status
AC-001	Registration number rejection for blank/whitespace input.	Client/Server test	Met
AC-002	First Name enforces minimum 3 characters and alphabetic content.	UI test	Met
AC-003	Last Name enforces minimum 3 characters and alphabetic content.	UI/API test	Met
AC-004	Email must follow the required email format.	UI/API test	Met
AC-005	Phone number must be numeric and exactly 10 digits.	UI/API test	Met
AC-006	New student password must not be empty.	UI test	Met
AC-007	New student password must be at least 8 characters.	UI test	Met
AC-008	Strong password composition must include upper, lower, number, and special character.	UI/API test	Met

ID	Acceptance Criterion	Verification Method	Status
AC-009	Invalid client-side values prevent submission and show an error.	Frontend verification	Met
AC-010	Backend invalid payloads return HTTP 400 Bad Request.	Direct API verification	Met
AC-011	Whitespace trimming is applied before validation.	Frontend/backend review	Met
AC-012	Valid student creation succeeds and the new record renders in the dashboard.	Manual UI test	Met
AC-013	Valid student update succeeds and the updated record renders in the dashboard.	Manual UI test	Met
AC-014	Invalid creation/update attempts are rejected without disrupting valid data.	Negative/regression test	Met
AC-015	Validation rules are consistently enforced for both Add and Edit operations.	Client/API test	Met
AC-016	Direct API requests cannot bypass validation and are rejected with 400.	API security test	Met
AC-017	Existing Edit and Delete actions still modify the correct student_id.	Regression/manual test	Met
AC-018	Existing administrator authentication and authorization remain unchanged.	Regression/security test	Met

4. Risk & Configuration Management

4.1 Risk Analysis

Risk	Impact	Likelihood	Mitigation / Control	Outcome
Incomplete Client Validation	Medium	Medium	Validate every required field before form submission and display formError feedback.	Mitigated
Client-Side Bypass	High	High	Duplicate every rule in studentController.js and reject invalid requests server-side.	Mitigated
Validation Inconsistency	High	Medium	Keep matching constraints across StudentFormModal.jsx and studentController.js.	Mitigated
Whitespace Circumvention	Medium	Medium	Trim all string-based inputs before checks are applied.	Mitigated
CRUD Regression	High	Low	Do not change studentModel.js, routing, or dashboard data-fetching logic; perform manual Edit/Delete verification.	Mitigated
Security Regression	High	Low	Preserve existing authentication, authorization, and password hashing mechanisms.	Mitigated
User Experience Confusion	Medium	Low	Use specific, user-facing error messages rather than generic submission failures.	Mitigated

4.2 Risk Mitigation Summary

The principal implementation risks are controlled through duplicated validation at the client and server layers, server-authoritative HTTP 400 rejection, and trimming before validation.

Regression risk is reduced by leaving database models, routes, and authentication middleware untouched and by explicitly testing existing Edit and Delete operations. The reported implementation identifies no outstanding implementation defects.

4.3 Configuration Items

CI ID	Configuration Item	Type	Changed?	Status
CI-01	StudentFormModal.jsx	Source Code	Yes	Modified
CI-02	studentController.js	Source Code / API	Yes	Modified
CI-03	studentModel.js	Source Code	No	Unchanged
CI-04	Student Dashboard CRUD Flow	Integration Behaviour	No	Regression Verified

CI ID	Configuration Item	Type	Changed?	Status
CI-05	CR-004 Implementation Report	Documentation	Yes	Prepared

4.4 Baseline and Version Information

Field	Details
Repository / Project	Not specified in the supplied CR-004 implementation report
Base Branch	Not specified in the supplied CR-004 implementation report
Starting Commit / Baseline ID	Not specified in the supplied CR-004 implementation report
Baseline Date	Not specified in the supplied CR-004 implementation report
Baseline Description	Existing student-management implementation prior to CR-004 validation changes
Working Branch	Not specified in the supplied CR-004 implementation report
Implementation Start Date	Not specified in the supplied CR-004 implementation report
Implementation Completion Date	08 October 2026

4.5 Commit Information

Commit ID	Commit Message	Purpose
Not specified	Not specified in the supplied CR-004 implementation report	Repository commit information was not included in the implementation source.
Working	CR-004 student input validation implementation	Represents the implementation scope described in the supplied report.

5. Implementation

5.1 Implementation Summary

CR-004 introduces comprehensive input validation for student Add and Edit operations. The frontend `StudentFormModal.jsx` validates input before submission and uses the existing `formError` state to display precise user-facing messages. The backend `studentController.js` independently re-validates each field so that direct or simulated requests cannot bypass the rules. All string-based inputs are trimmed before validation. Existing CRUD operations, authentication, authorization, and password hashing are preserved.

5.2 Technical Implementation

Technical Area	Changed?	Implementation Details
Frontend	Yes	Added client-side validation in <code>StudentFormModal.jsx</code> , including field rules, trimming, submission prevention, and precise error messages.
Backend	Yes	Added independent validation in <code>addStudent</code> and <code>editStudent</code> ; failed validation returns HTTP 400 before DB work.
Database	No	No schema redesign or <code>studentModel.js</code> SQL change required.
API	Yes	Existing student Add/Edit endpoints now enforce validation and reject invalid request bodies with descriptive 400 responses.
Authentication / Authorization	No	Existing administrator authentication and authorization behaviour remains unchanged.
Security	Yes	Server-side validation prevents client-bypass; trimming prevents whitespace circumvention.
Configuration	No	No environment, dependency, or deployment configuration change is specified.
External Services	No	No external service is affected.

5.3 Files Modified

File	Component	Purpose of Modification
<code>frontend/src/components/StudentFormModal.jsx</code>	React Component	Client-side validation for Add/Edit form submissions, trimming, and user-facing <code>formError</code> messages.
<code>backend/src/controllers/studentController.js</code>	REST Controller	Server-side validation for <code>addStudent</code> and <code>editStudent</code> ; HTTP 400 rejection before database operations.

Unmodified but deliberately preserved: backend/src/models/studentModel.js; routing mechanisms; dashboard data-fetching pipelines; authentication/authorization middleware; password hashing.

5.4 Database Changes

Field	Details
Database Changes	No
Schema Changes	None
Tables / Columns	No changes
Migrations	Not required
Model SQL Changes	None - studentModel.js remains unchanged
Data Integrity Approach	Reject invalid requests before database operations; valid requests continue through existing CRUD logic.

5.5 API Changes

Method / Function	Change	Purpose
addStudent	Validation added	Reject invalid creation payloads and return HTTP 400 before DB operations.
editStudent	Validation added	Reject invalid update payloads and return HTTP 400 before DB operations.
Existing student response	No structural redesign	Existing response mechanism remains; validation only changes which requests are accepted.

5.6 Authentication / Authorization / Security Changes

Authentication and authorization mechanisms were preserved. The change adds validation controls rather than introducing a new access-control model. The security focus is on server-side validation, direct-request rejection, trimming, and preservation of existing student data isolation and administrator controls.

5.7 Implementation Logic

Main Workflow: Student Add / Edit with Dual-Layer Validation

1. The administrator enters student information in StudentFormModal.jsx.
2. Client-side validation trims string inputs and checks required values, field formats, and password strength where applicable.
3. When client validation fails, form submission is stopped and the formError state displays the specific validation message.
4. When client validation succeeds, the request proceeds to the existing student Add or Edit API operation.
5. The backend controller trims/normalizes string inputs and independently verifies the same validation rules.
6. If backend validation fails, the controller returns HTTP 400 Bad Request with a descriptive message and does not proceed to database operations.
7. If validation succeeds, existing CRUD logic continues without changes to student identifiers or password hashing.

5.8 Error and Data Handling

Handling Area	Implementation
Client Error Handling	Specific validation messages are shown through the existing formError UI state and invalid submission is stopped before an API call.
Server Error Handling	Invalid requests return HTTP 400 Bad Request with descriptive JSON error feedback.
Whitespace Handling	All string-based inputs are trimmed before validation, preventing padding-based circumvention.
Password Handling	Strong password requirements apply to new student creation; existing hashing behaviour is preserved.
Persistence Handling	Invalid requests do not proceed to database operations; valid requests use the existing CRUD persistence path.

5.9 Important Technical Decisions

- Dual-layer validation: frontend checks improve usability, while backend checks remain authoritative for security.
- Controller-level enforcement and preservation of existing CRUD/security mechanisms keep the change focused on validation while avoiding unnecessary model and access-control changes.

6. Testing & Verification

6.1 Testing Strategy

Applicable testing types:

- Functional testing
- Positive testing
- Negative testing
- Boundary testing
- Validation testing
- API testing
- Frontend testing
- Backend testing
- Authentication testing
- Authorization testing
- Security testing
- Regression testing
- Error-handling testing
- Null / missing-data testing
- Usability testing
- Compatibility testing

6.2 Testing Environment

Field	Details
Environment	Local development / test environment
Application Version	Version 1.0
Browser / Client	Not specified in the supplied implementation report
Database	Existing project database; specific engine/version not specified
API / Server Environment	Existing Express/Node.js backend; exact version not specified

6.3 Test Data / Test Account

Field	Details
Test Data	Examples recorded in the implementation report: "Jo", "John123", "john.doe.com", "test@", 9/11-digit phone values, "password", and "Password123".

Field	Details
Valid Data	Properly filled student form data satisfying all required rules.
Authenticated Context	Administrator/student management context; exact test account identifiers were not specified.
Permissions	Existing administrator authentication and authorization behaviour preserved.

6.4 Test Cases

Test ID	Req / AC	Test Case	Input / Action	Expected Result	Status
TC-01	AC-001	Blank First Name	Submit with blank First Name	Prevent submission and show validation error	Pass
TC-02	AC-002	First Name < 3 chars	First Name = "Jo"	Prevent submission and show validation error	Pass
TC-03	AC-002	Name with numbers	First Name = "John123"	Prevent submission and show validation error	Pass
TC-04	AC-004	Invalid email format	Email = "john.doe.com"	Prevent submission and show validation error	Pass
TC-05	AC-005	Phone < 10 digits	Phone = "123456789"	Prevent submission and show validation error	Pass
TC-06	AC-005	Phone > 10 digits	Phone = "12345678901"	Prevent submission and show validation error	Pass
TC-07	AC-005	Phone contains letters	Phone = "1234567abc"	Prevent submission and show validation error	Pass
TC-08	AC-008	Weak password	Password = "password"	Prevent submission and show validation error	Pass
TC-09	AC-012	Valid Student Creation	Submit properly filled Add Student form	API succeeds and new student renders in dashboard	Pass
TC-10	AC-013	Valid Student Update	Submit valid Edit Student changes	Updated values render correctly in dashboard	Pass
TC-11	AC-010	Missing First Name API	Simulated payload with missing firstName	HTTP 400 Bad Request	Pass
TC-12	AC-003	Short Last Name API	lastName = "Li"	HTTP 400 Bad Request	Pass

Test ID	Req / AC	Test Case	Input / Action	Expected Result	Status
TC-13	AC-002	Numeric First Name API	firstName = "A1"	HTTP 400 Bad Request	Pass
TC-14	AC-004	Bad Email API	email = "test@"	HTTP 400 Bad Request	Pass
TC-15	AC-005	Short Phone API	phoneNumber = "123"	HTTP 400 Bad Request	Pass
TC-16	AC-008	Password without special char	password = "Password123"	HTTP 400 Bad Request	Pass
TC-17	AC-017	Existing Edit/Delete targeting	Manually edit/delete an existing student	Correct student_id is targeted and updated/deleted	Pass
TC-18	AC-018	Existing security controls	Use existing administrator authentication/authorization	Existing access-control behaviour unchanged	Pass

6.5 Test Summary

Field	Details
Total Test Cases	18
Passed	18
Failed	0
Blocked	0
Not Tested	0
Pass Rate	100 %
Overall Test Result	Passed

Summary: All client-side, server-side, CRUD, and security verification scenarios recorded in the supplied implementation report passed. No implementation defects or regressions were identified in the stated test results.

6.6 Regression Verification

Regression testing focused on preserving existing Edit and Delete behaviour, student identifiers, CRUD flow, authentication, authorization, and password hashing. The implementation report explicitly records that Edit and Delete continue to target the correct student_id and that the underlying model, routing, dashboard data-fetching, authentication, authorization, and password hashing mechanisms were preserved.

6.7 Acceptance Criteria Verification

Acceptance Criterion	Test / Verification	Result	Evidence Reference
AC-001 - AC-011	TC-01 - TC-08, TC-11 - TC-16	Met	Evidence 01 / 04
AC-012 - AC-015	TC-09 - TC-10 plus invalid Add/Edit scenarios	Met	Evidence 02 / 03
AC-016	TC-11 - TC-16 direct API checks	Met	Evidence 04
AC-017	TC-17 manual regression verification	Met	Evidence 05
AC-018	TC-18 security regression verification	Met	Existing security controls

7. Traceability & Evidence

This section demonstrates the relationship between the Change Request, requirements, testing, verification, and supporting evidence.

7.1 Change Traceability

Requirement / AC	Test Case(s)	Evidence ID	Status
FR-01 - FR-05	TC-01 - TC-07, TC-11 - TC-15	Evidence 01 / 04	Met
FR-06 - FR-10	TC-08, TC-16 and password checks	Evidence 01 / 04	Met
FR-11 - FR-15	TC-01 - TC-10	Evidence 01 - 03	Met
FR-16	TC-17	Evidence 05	Met
FR-17 - FR-18	TC-18, regression review	Evidence 05 / existing controls	Met
NFR-01 - NFR-04	TC-01 - TC-18	Evidence 01 - 05	Met
SEC-001 - SEC-006	TC-11 - TC-18	Evidence 04 / 05	Met
AC-001 - AC-018	TC-01 - TC-18	Evidence 01 - 05	Met

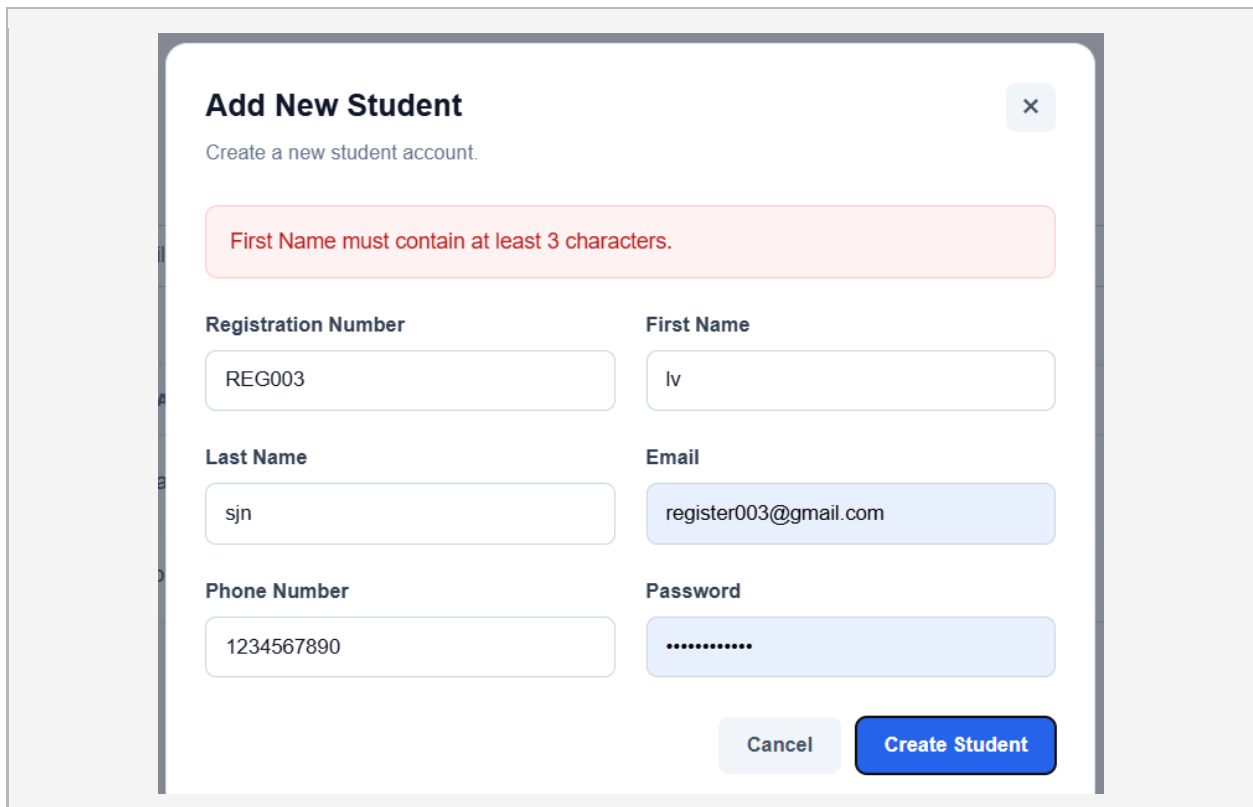
Traceability Chain: Change Request -> Requirement -> Test -> Evidence -> Verification

7.2 Embedded Evidence

Provide relevant screenshots, application outputs, API responses, manual test results, or other supporting evidence. The actual screenshots were not included in the current source package, so the evidence locations below are prepared as insertion points for the screenshots referenced in the CR-004 request.

Evidence-01

Related Requirement / Test: AC-002 / TC-02 - StudentFormModal displays the specific validation message for a first name shorter than 3 characters.

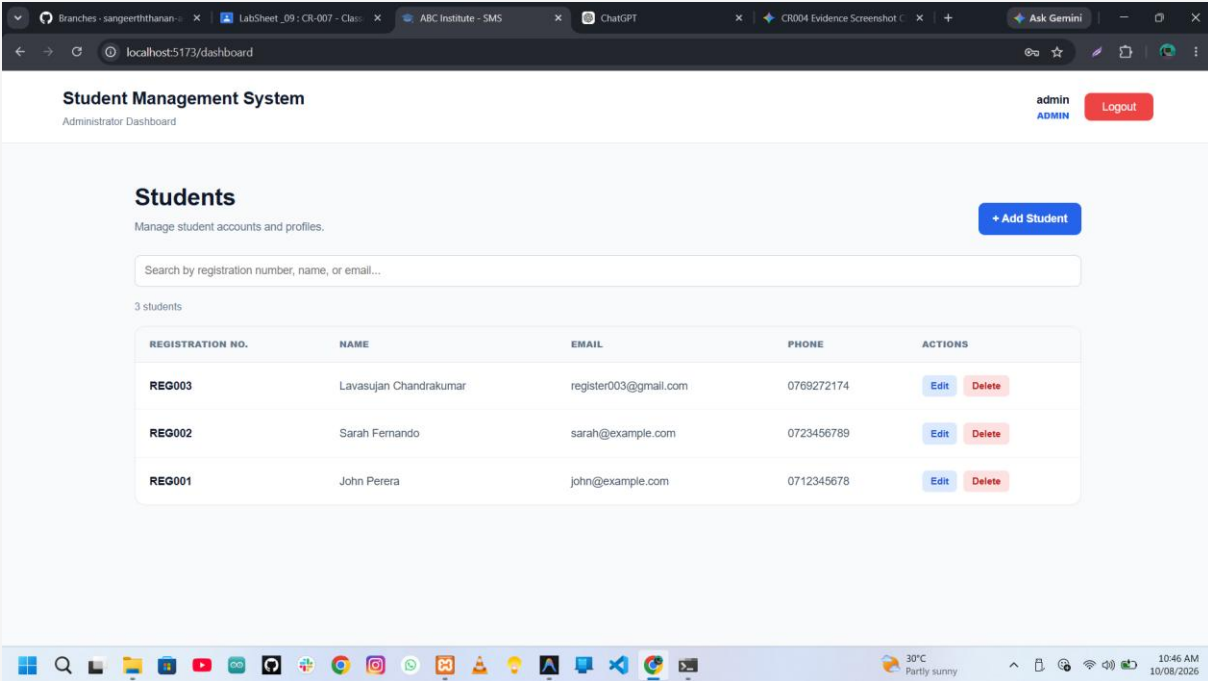


The screenshot displays a modal window titled "Add New Student" with a close button (X) in the top right corner. Below the title is the subtitle "Create a new student account." A red validation message box at the top states: "First Name must contain at least 3 characters." The form contains six input fields arranged in two columns. The left column includes "Registration Number" (containing "REG003"), "Last Name" (containing "sjn"), and "Phone Number" (containing "1234567890"). The right column includes "First Name" (containing "lv"), "Email" (containing "register003@gmail.com"), and "Password" (containing masked characters "....."). At the bottom right, there are two buttons: a light blue "Cancel" button and a dark blue "Create Student" button.

Figure 1: StudentFormModal validation error showing the minimum first-name length requirement

Evidence-02

Related Requirement / Test: AC-012 / TC-09 - Valid student creation succeeds and the new student record is rendered in the dashboard table.



The screenshot displays the 'Student Management System' Administrator Dashboard. The page title is 'Students' with the subtitle 'Manage student accounts and profiles.' A '+ Add Student' button is located in the top right corner. Below the title is a search bar with the placeholder text 'Search by registration number, name, or email...'. A message indicates '3 students' are listed. The table below contains the following data:

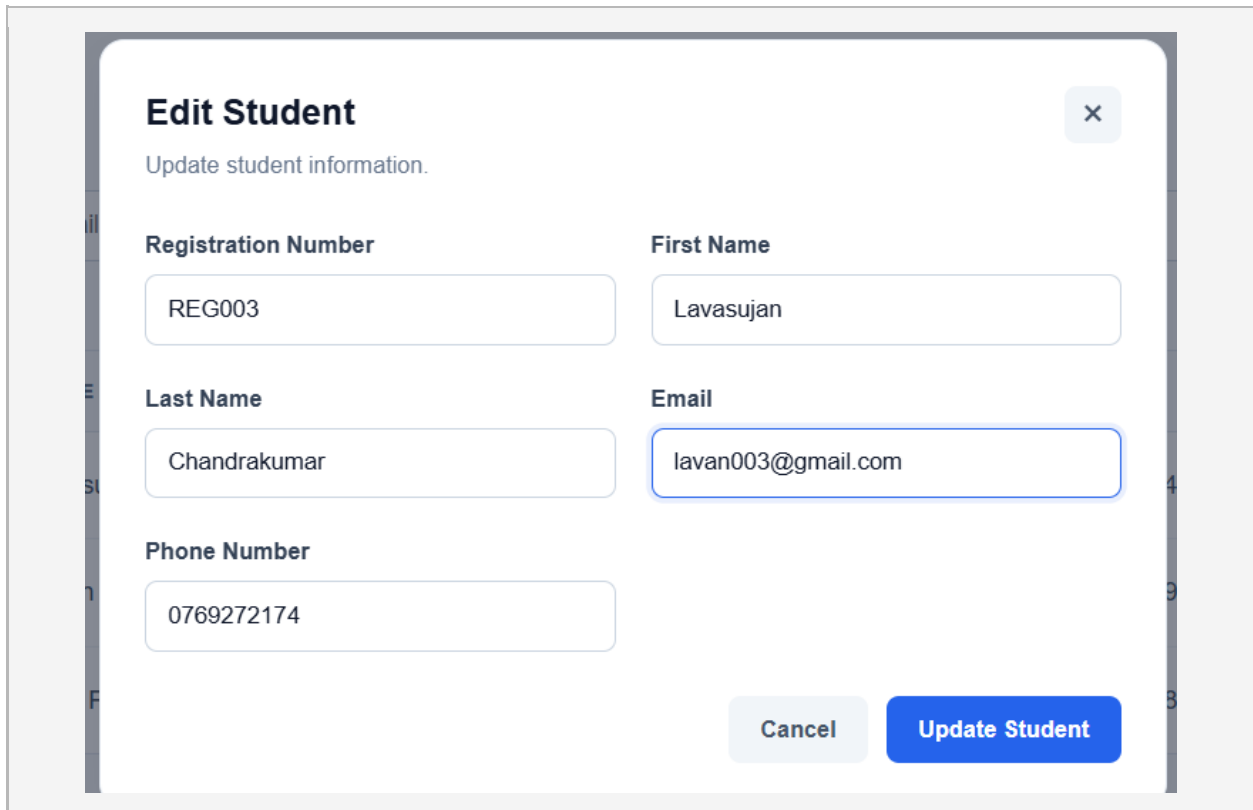
REGISTRATION NO.	NAME	EMAIL	PHONE	ACTIONS
REG003	Lavasujan Chandrakumar	register003@gmail.com	0769272174	Edit Delete
REG002	Sarah Fernando	sarah@example.com	0723456789	Edit Delete
REG001	John Perera	john@example.com	0712345678	Edit Delete

The dashboard also shows the user 'admin' with a 'Logout' button in the top right corner. The browser's address bar shows 'localhost:5173/dashboard'. The Windows taskbar at the bottom shows the time as 10:46 AM on 10/08/2026, with a weather forecast of 30°C Partly sunny.

Figure 2: Valid student creation - successful modal submission and dashboard record

Evidence-03

Related Requirement / Test: AC-013 / TC-10 - Valid student update succeeds and the changed values are visible in the dashboard list.



The screenshot displays a web application interface for editing student information. The form is titled "Edit Student" and includes a subtitle "Update student information." with a close button (X) in the top right corner. The form contains five input fields arranged in a grid: "Registration Number" (REG003), "First Name" (Lavasujan), "Last Name" (Chandrakumar), "Email" (lavan003@gmail.com), and "Phone Number" (0769272174). The "Email" field is highlighted with a blue border. At the bottom right, there are two buttons: "Cancel" and "Update Student".

Field Label	Value
Registration Number	REG003
First Name	Lavasujan
Last Name	Chandrakumar
Email	lavan003@gmail.com
Phone Number	0769272174

Figure 3: Valid student update - edited values reflected in the dashboard

Evidence-04

Related Requirement / Test: AC-010, AC-016 / TC-11 - TC-16 - Backend rejects invalid direct requests with HTTP 400.

Example response for an invalid email request:

```
{
  "success": false,
  "message": "Email must follow a valid email format."
}
```

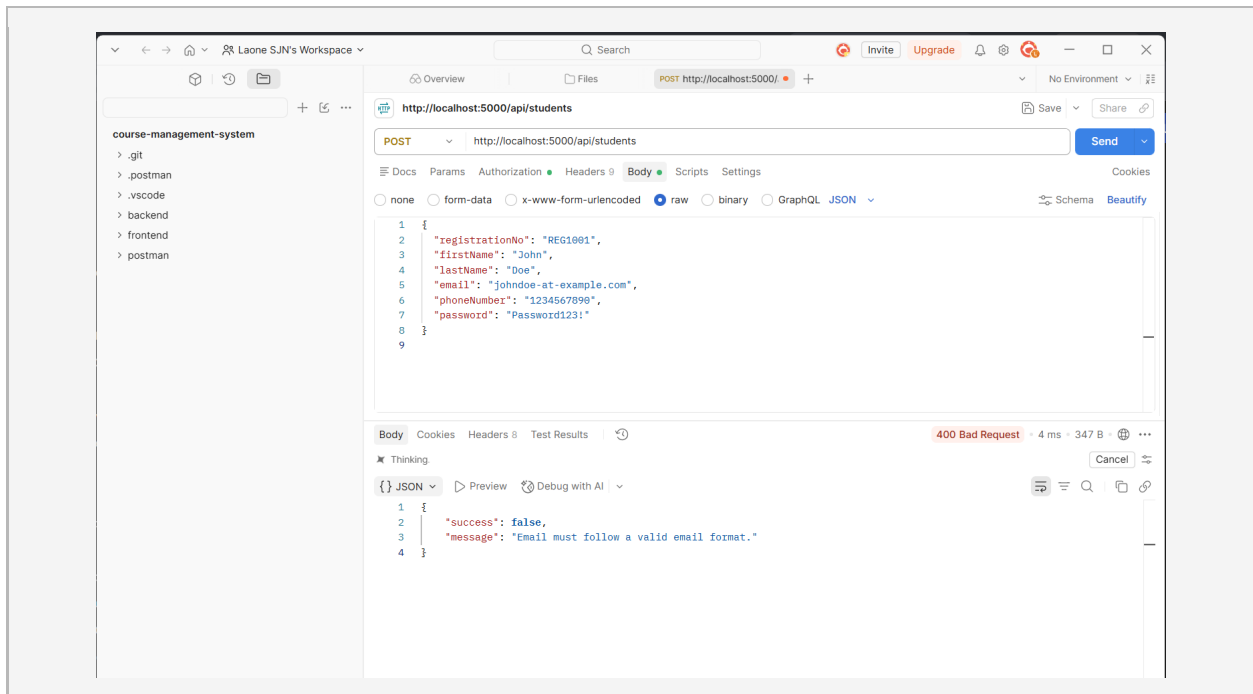


Figure 4: Backend rejection of an invalid student request with HTTP 400 Bad Request

Evidence-05

Related Requirement / Test: AC-017 / TC-17 - Existing Edit and Delete actions continue to affect the correct student_id.

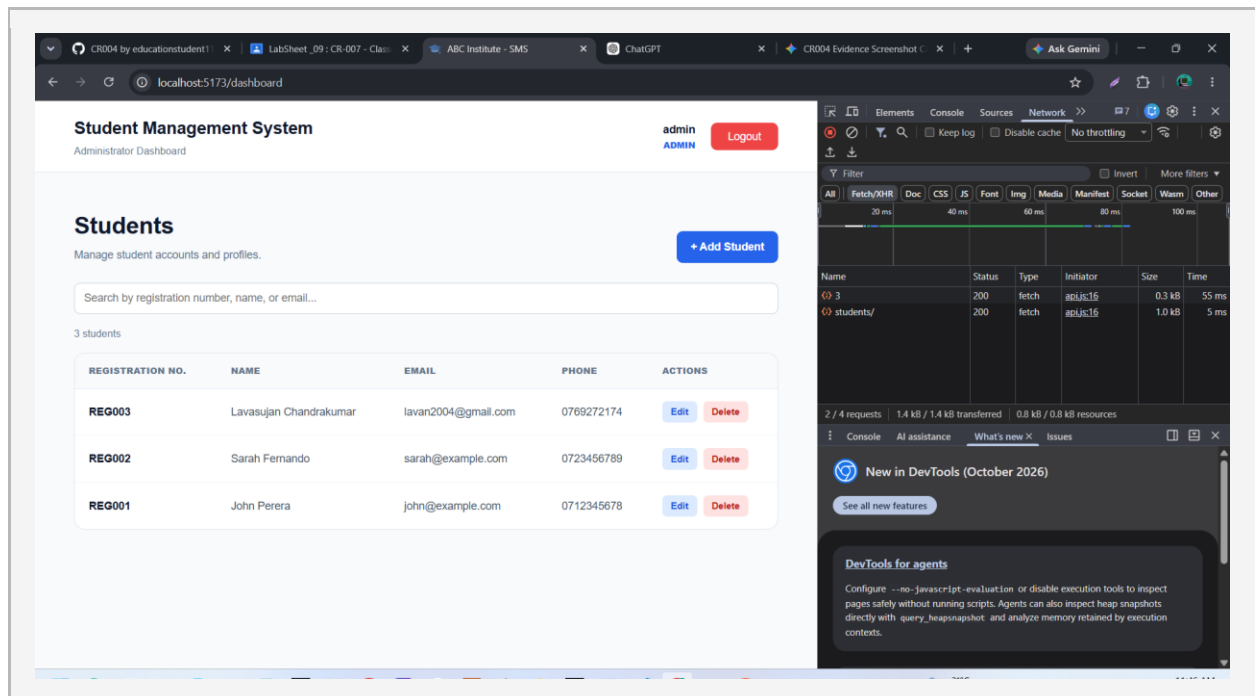


Figure 5: Existing Edit and Delete functionality targeting the correct student_id

8. Version Control & Change Control

8.1 Repository and Branch Information

Field	Details
Repository	Not specified in the supplied CR-004 implementation report
Source Branch	Not specified in the supplied CR-004 implementation report
Target Branch	Not specified in the supplied CR-004 implementation report
Pull Request ID	Not specified in the supplied CR-004 implementation report
Pull Request Date	Not specified in the supplied CR-004 implementation report
Reviewer	Not specified in the supplied CR-004 implementation report
Review Status	Not specified in the supplied CR-004 implementation report
Merge Status	Not specified in the supplied CR-004 implementation report

8.2 Code Review / Change Review

Review Comment	Action Taken	Commit ID	Status
Client-side validation coverage	StudentFormModal.jsx validation rules and formError feedback implemented.	Not specified	Ready for review
Server-side validation coverage	studentController.js re-validates inputs and returns HTTP 400 for invalid requests.	Not specified	Ready for review
Regression / CRUD preservation	studentModel.js, routing, dashboard data-fetching, and existing CRUD behaviour preserved and tested.	Not specified	Verified
Security controls	Backend validation and trimming reduce bypass opportunities while existing authentication/authorization remain intact.	Not specified	Verified

8.3 Change Control Outcome

The supplied implementation report records the implementation as complete and ready for deployment. Formal repository, branch, Pull Request, and reviewer metadata were not included in the source material and therefore remain marked as not specified rather than being inferred.

9. Final Status & Closure

9.1 Final Change Status

Field	Details
Change Request ID	CR-004
Change Request Title	Student Input Field Validation
Implementation Status	Completed
Testing Status	Passed
Acceptance Criteria	All Met
Demonstration / Review	Evidence prepared; actual screenshots pending insertion
Final Acceptance	Ready for formal review / submission
Final Change Status	Implementation Complete / Ready for Deployment

9.2 Change Scope Summary

Area	Changed?
Frontend	Yes
Backend	Yes
Database	No
API	Yes - validation behaviour
Authentication / Authorization	No - preserved
Security	Yes
Configuration	No
External Services	No
Documentation	Yes

9.3 Final Summary

Change Request CR-004 implements dual-layer validation for student Add and Edit operations. StudentFormModal.jsx provides immediate client-side validation and clear error messages, while studentController.js independently enforces the same constraints and rejects invalid direct requests with HTTP 400. Input trimming prevents whitespace circumvention, strong password validation is applied to new student creation, and existing CRUD, authentication, authorization, and password hashing mechanisms remain preserved. The implementation source reports all stated tests and acceptance checks as passed.

9.4 Outstanding Issues / Limitations

- No implementation defects were identified in the supplied source report.
- Actual screenshot files are not present in the current report package and must be inserted into the prepared evidence placeholders before final submission.
- Repository, branch, commit, Pull Request, and reviewer details were not specified in the supplied implementation report and have therefore not been invented.

9.5 Follow-up Actions

- Insert the five screenshot evidence items into Evidence-01 to Evidence-05.
- Replace repository / branch / commit / review placeholders when the authoritative version-control details are available.
- Perform final document review before submission.

9.6 Closure Checklist

Closure Item	Status
Change requirements reviewed	Completed
Change analysis completed	Completed
Impact analysis completed	Completed
Assumptions and constraints documented	Completed
Risk analysis completed	Completed
Configuration items identified	Completed
Baseline identified	Documented as not specified where source data was unavailable
Implementation completed	Completed
Test cases completed	Completed
Acceptance criteria verified	Completed
Evidence locations prepared	Completed

Closure Item	Status
Traceability completed	Completed
Version control information recorded	Placeholders retained where not specified
Review / Pull Request completed	Not specified in source
Outstanding issues recorded	Completed
Follow-up actions recorded	Completed
Report reviewed for completeness	Prepared for final evidence insertion
Change ready for closure	Yes - pending final evidence and review metadata

9.7 Closure Comments

The validation implementation provides stronger student-data integrity and security without requiring database redesign. The implementation is focused, backward-compatible, and supported by client-side, server-side, and regression verification described in the supplied report.

9.8 Final Closure

Final Change Status: Implementation Complete / Ready for Deployment

Closed By: C. Lavasujan (Student ID: 24ICT06)

Closure Date: 08 October 2026

Confirmation: I confirm that the information provided in this report accurately represents the implementation, testing, verification, and current status of Change Request CR-004, subject to the evidence and version-control placeholders explicitly identified in this document.

Student Name: C. Lavasujan

Signature: _____

Date: 08 October 2026

Appendix A - Additional Evidence

Evidence A-01

Description: Client-side validation rule coverage for required student fields, names, email, phone, and new-student password.

Related Requirement / Test: FR-01 - FR-12 / TC-01 - TC-08, TC-11 - TC-16

```
Validation rule summary:  
First Name >= 3 chars and alphabetic  
Last Name >= 3 chars and alphabetic  
Email matches required format  
Phone = 10 numeric digits  
Password >= 8 chars + upper + lower + number + special
```

Evidence A-02

Description: Server-side rejection behaviour for invalid requests. The implementation report records 400 Bad Request for invalid payloads including bad email, short phone, missing first name, short last name, numeric first name, and a password without a special character.

```
HTTP 400 Bad Request  
{  
  "success": false,  
  "message": "Email must follow a valid email format."  
}
```

Appendix B - Screenshot Evidence

B-01 - Validation Error

StudentFormModal showing "First Name must contain at least 3 characters."

The screenshot shows a modal window titled "Add New Student" with a close button (X) in the top right corner. Below the title is the instruction "Create a new student account." A red error message box at the top states: "First Name must contain at least 3 characters." The form contains six input fields arranged in two columns. The left column has fields for "Registration Number" (containing "REG003"), "Last Name" (containing "sjn"), and "Phone Number" (containing "1234567890"). The right column has fields for "First Name" (containing "lv"), "Email" (containing "register003@gmail.com"), and "Password" (containing masked characters "....."). At the bottom right, there are two buttons: a light blue "Cancel" button and a dark blue "Create Student" button.

Field Label	Value
Registration Number	REG003
First Name	lv
Last Name	sjn
Email	register003@gmail.com
Phone Number	1234567890
Password	

Figure 1: StudentFormModal showing "First Name must contain at least 3 characters."

B-02 - Valid Student Creation

Successful Add Student submission and the new dashboard record

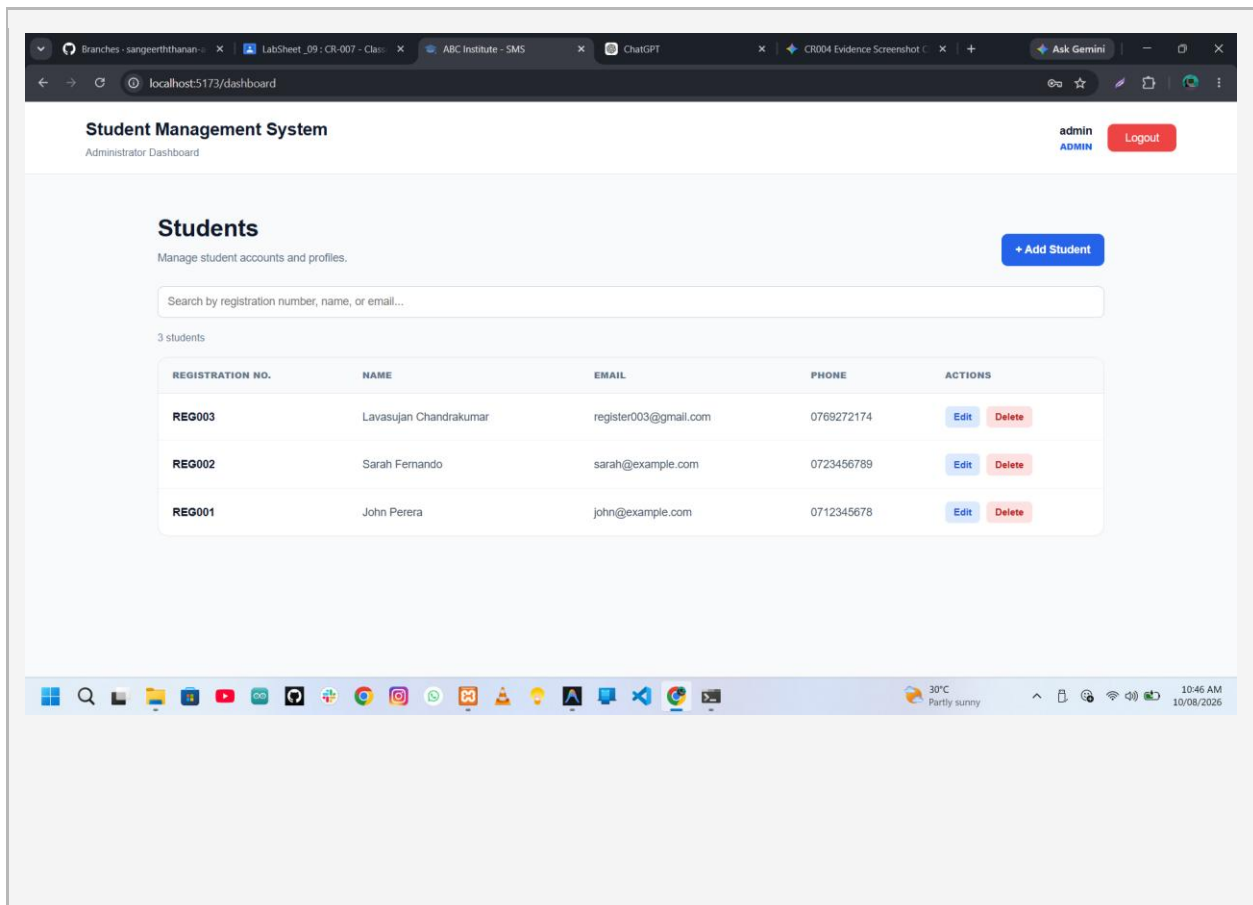


Figure 2: Successful Add Student submission and the new dashboard record

B-03 - Valid Student Update

Successful Edit submission and updated dashboard record

The screenshot displays the 'Student Management System' Administrator Dashboard. The page title is 'Students' with the subtitle 'Manage student accounts and profiles.' A '+ Add Student' button is located in the top right corner. A search bar is present with the placeholder text 'Search by registration number, name, or email...'. Below the search bar, it indicates '3 students' are listed. The table below shows the student records:

REGISTRATION NO.	NAME	EMAIL	PHONE	ACTIONS
REG003	Lavasujan Chandrakumar	lavan003@gmail.com	0769272174	Edit Delete
REG002	Sarah Fernando	sarah@example.com	0723456789	Edit Delete
REG001	John Perera	john@example.com	0712345678	Edit Delete

The dashboard is viewed through a web browser with multiple tabs open. The system clock at the bottom right shows 10:47 AM on 10/08/2026, with a weather forecast of 30°C Partly sunny.

Figure 3: Successful Edit submission and updated dashboard record

B-04 - Backend Rejection

HTTP 400 Bad Request response for an invalid API request

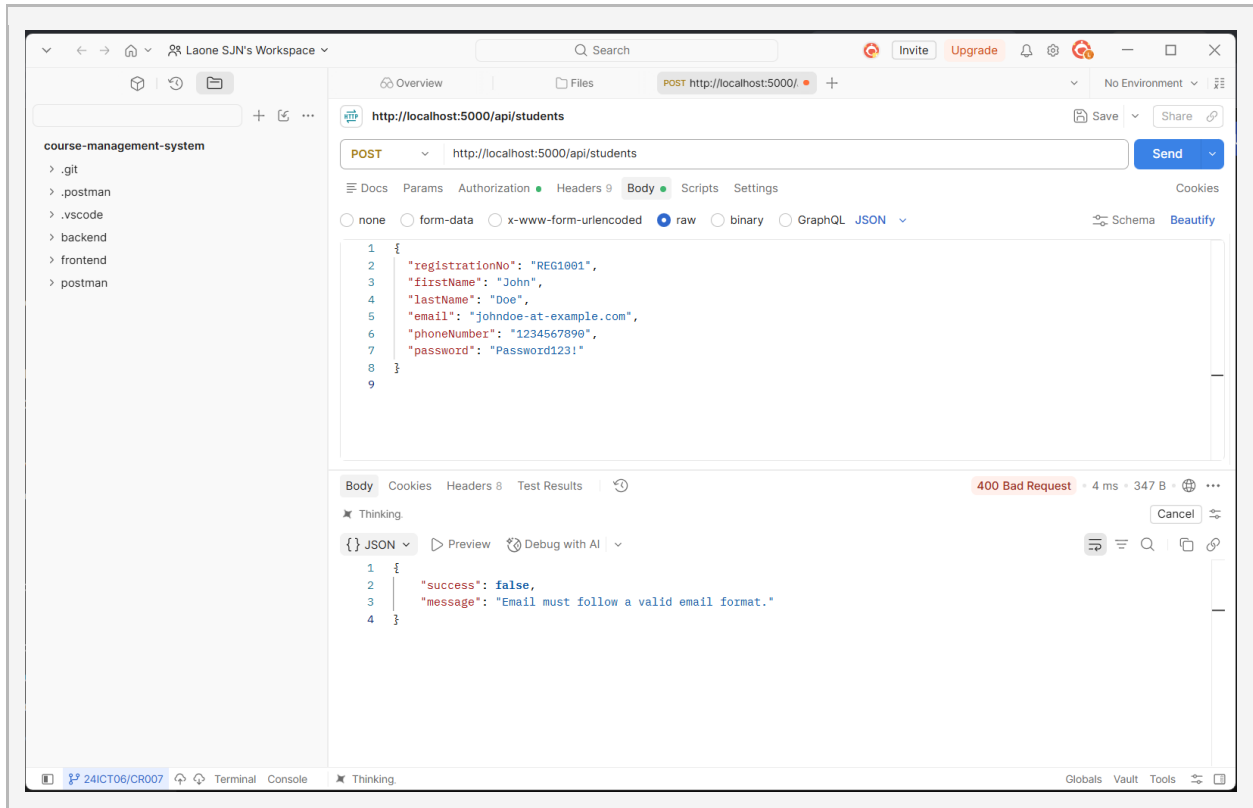


Figure 4: HTTP 400 Bad Request response for an invalid API request

Appendix C - Original Change Request Reference

Original Change Request: CR-004

Field	Details
Change Request ID	CR-004
Change Request Title	Student Input Field Validation
Prepared By	C. Lavasujan
Registration Number	24ICT06
Reference Basis	Supplied implementation report: client-side and server-side validation for StudentFormModal.jsx and studentController.js.

Evidence Requirements Referenced by the Change Request

- Screenshot of StudentFormModal displaying "First Name must contain at least 3 characters."
- Screenshot showing successful student creation followed by the new dashboard record.
- Screenshot showing successful student update reflected in the dashboard list.
- API client / response-log evidence showing HTTP 400 Bad Request for an invalid backend request.
- Manual testing evidence confirming existing Edit and Delete actions continue to target the correct student_id.

STUDENT REPORT COMPLETION CHECK

Completion Item	Status
Original Change Request has been reviewed.	Completed
Business requirement has been documented.	Completed
Current behaviour has been documented.	Completed
Requested change has been documented.	Completed
Scope has been identified.	Completed
Assumptions and constraints have been documented.	Completed
Impact analysis has been completed.	Completed
Affected components have been identified.	Completed
Affected files have been identified.	Completed
Configuration items have been recorded.	Completed
Baseline has been recorded / unavailable fields flagged.	Completed
Version-control information has been recorded / unavailable fields flagged.	Completed
Risks and mitigations have been documented.	Completed
Requirements have been mapped to implementation.	Completed
Implementation details have been documented.	Completed
Technical logic has been explained.	Completed
Error handling has been documented.	Completed
Null / missing-data handling has been documented.	Completed
Testing strategy has been defined.	Completed
Test cases have been completed.	Completed
Test results have been recorded.	Completed
Acceptance criteria have been verified / evidence prepared.	Completed
Requirement traceability has been completed.	Completed
Evidence locations have been prepared.	Completed
Version-control / Pull Request information has been prepared for completion.	Completed
Regression verification has been addressed.	Completed
Demonstration / review has been prepared for completion.	Completed

Completion Item	Status
Outstanding issues have been recorded.	Completed
Follow-up actions have been recorded.	Completed
Final change status has been determined.	Completed
Closure checklist has been completed.	Completed